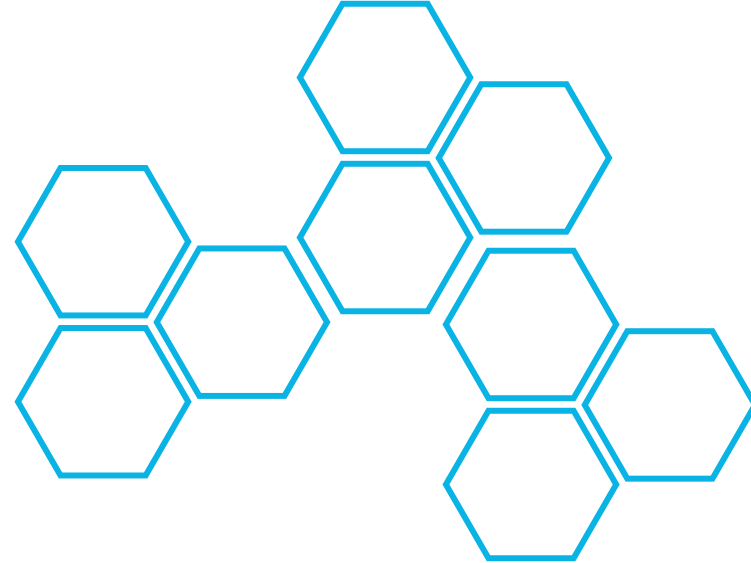
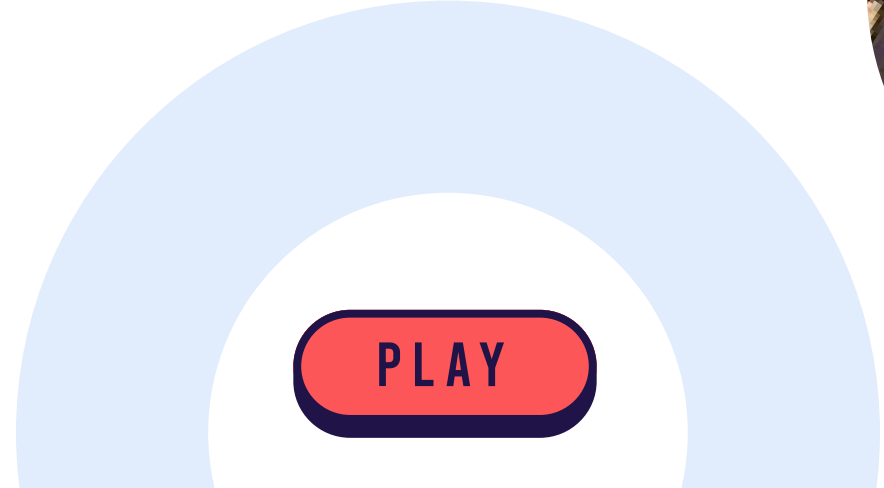


Play!! Now!



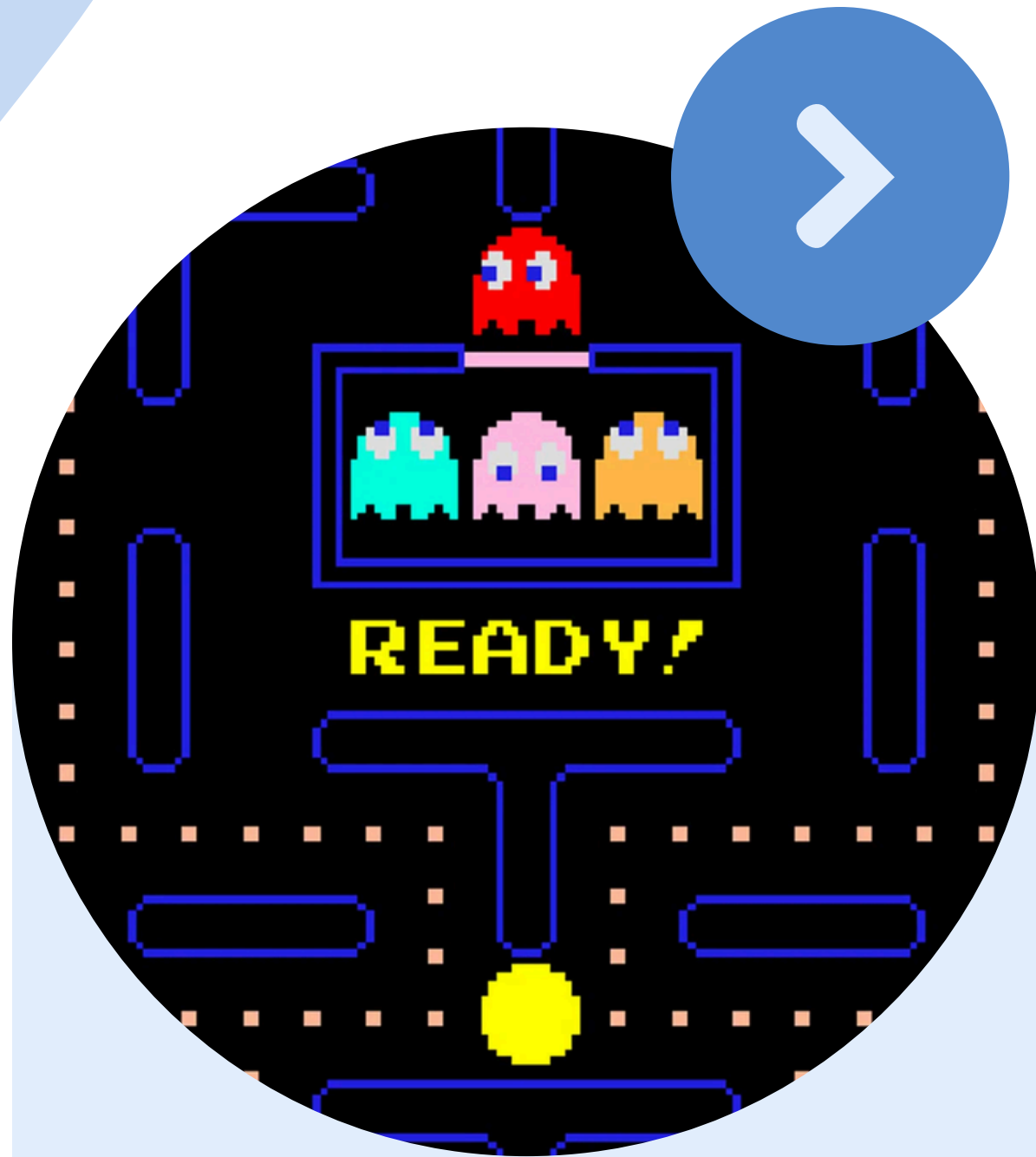
EXPLORING THE WUMPUS WORLD

A Logical Agent Adventure



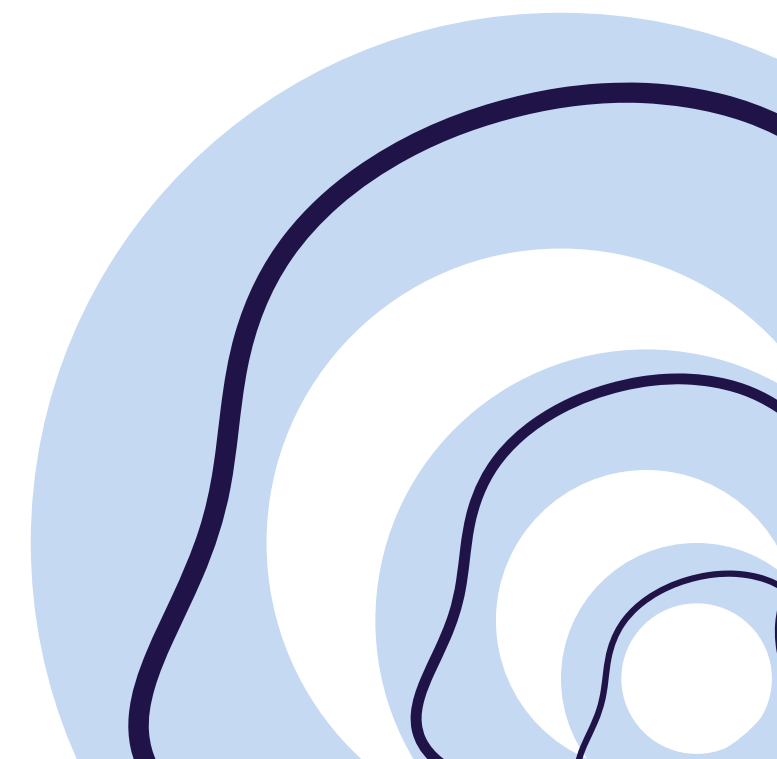
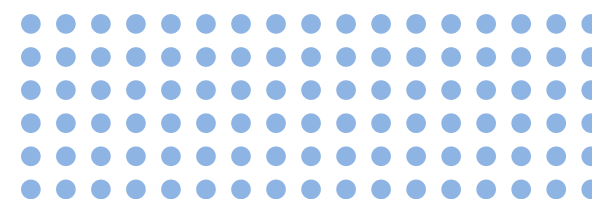
PLAY





Abstract

The Wumpus World is a classic problem in artificial intelligence, demonstrating how agents can make decisions under uncertainty. It is a grid-based environment where a rational agent must find gold while avoiding hidden dangers like pits and the Wumpus (a monster). This project explores the modeling of the environment, the agent's knowledge base, inference mechanisms, and decision-making strategies to safely and efficiently navigate the world.



Problem Statement



➤ In the Wumpus World, an intelligent agent must achieve two main objectives:

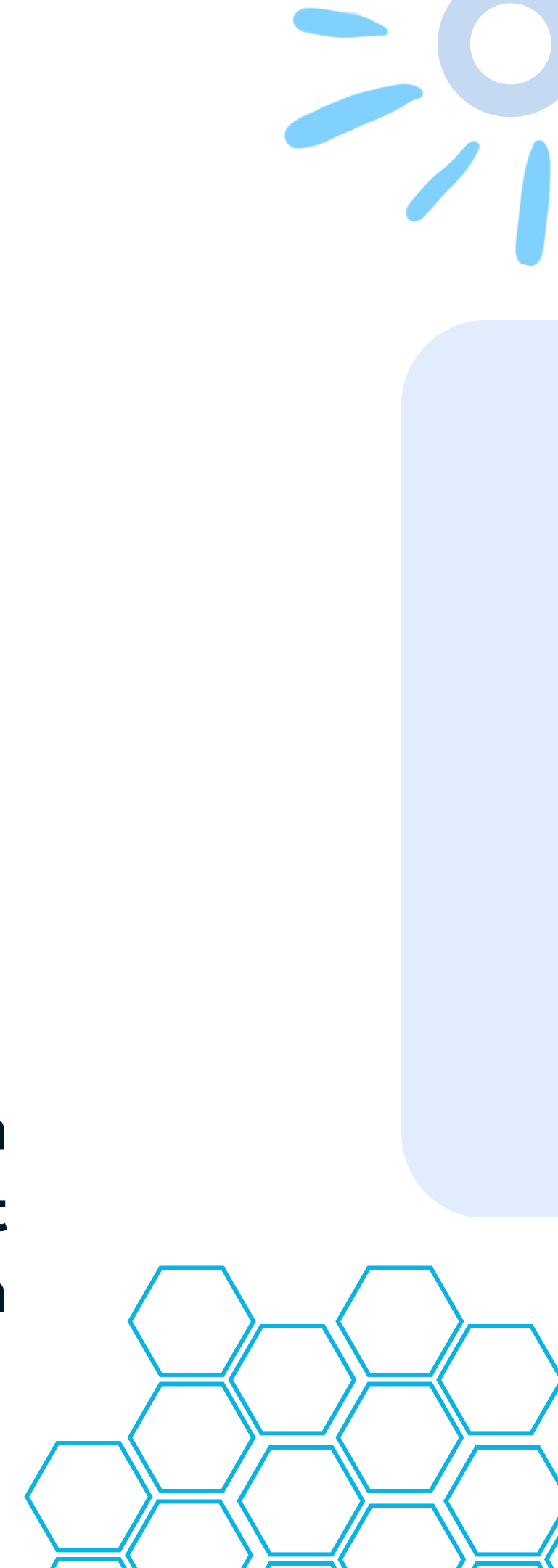


Locate and retrieve the gold hidden somewhere in the environment.

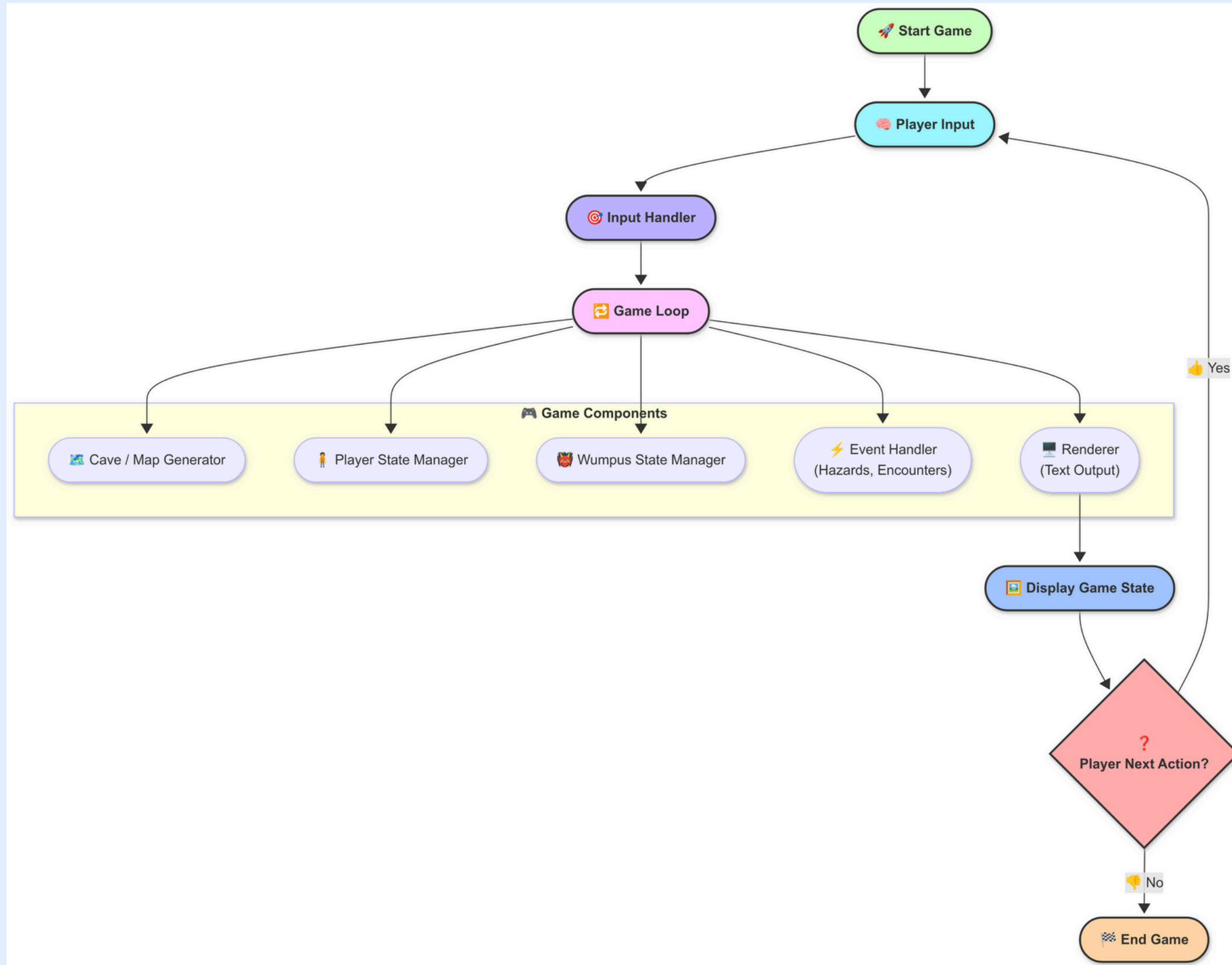


Avoid deadly hazards such as pits and the Wumpus.

The challenge lies in making decisions based on partial, uncertain information using percepts (breeze, stench, Gold, etc.). The agent must reason logically to deduce the safest moves and plan a successful strategy. >>>>>



Architecture Diagram





Methodology



Environment Design

A 4x4 or 5x5 grid with random placement of Wumpus, pits, and gold



Decision Making

- If gold percept detected (glitter) → grab.
- If stench/breeze → mark danger zones.
- If safe → move forward.

Agent Design

- Uses Percepts to update a Knowledge Base.
- Applies Logical Inference to reason about safe cells.
- Applying Backtracking

Implementation Details

Language/Tools Used:



Libraries Used in Python:



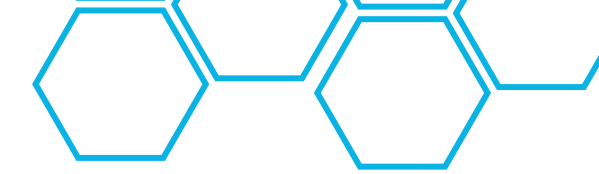
Python Enum

Queues

Game Rules:

- **User Movement**
 - Agent can move, grab
- **User Skill**
 - Agent can Sense pits through breeze int the adjacent block
 - Agent can Sense wumpus through stretch int the adjacent block
- **User Objective**
 - Objective is to maximize score (gold & survival).
- **User Survival Instincts**
 - Agent "dies" if it falls into a pit or Wumpus eats it.

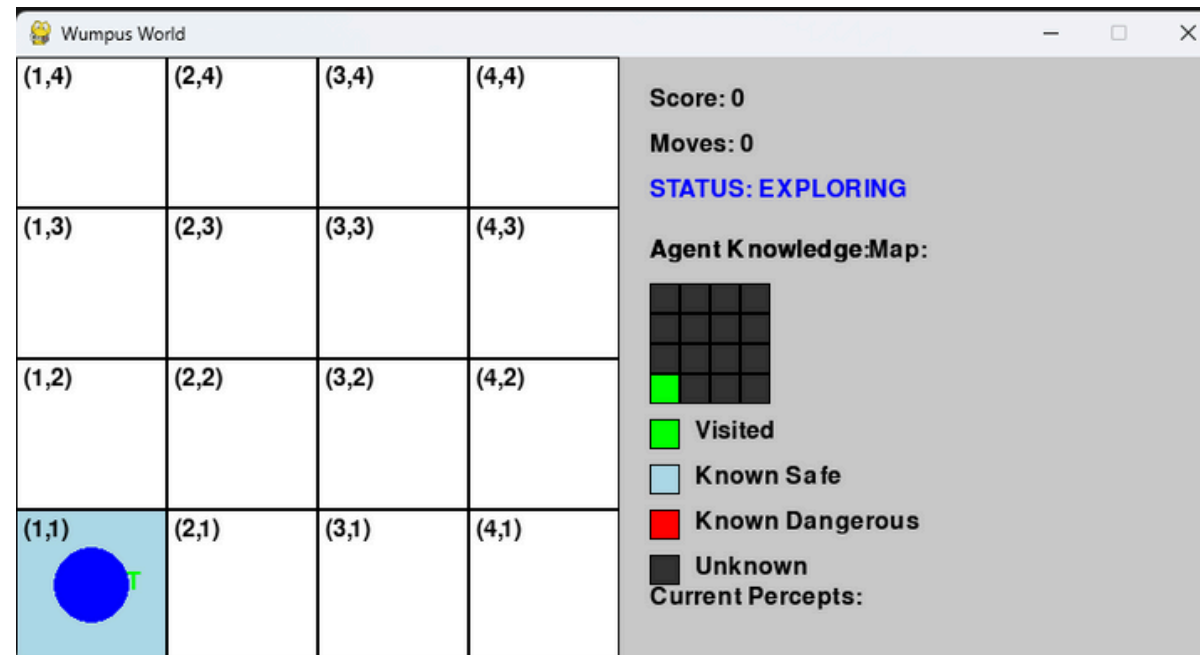
Results



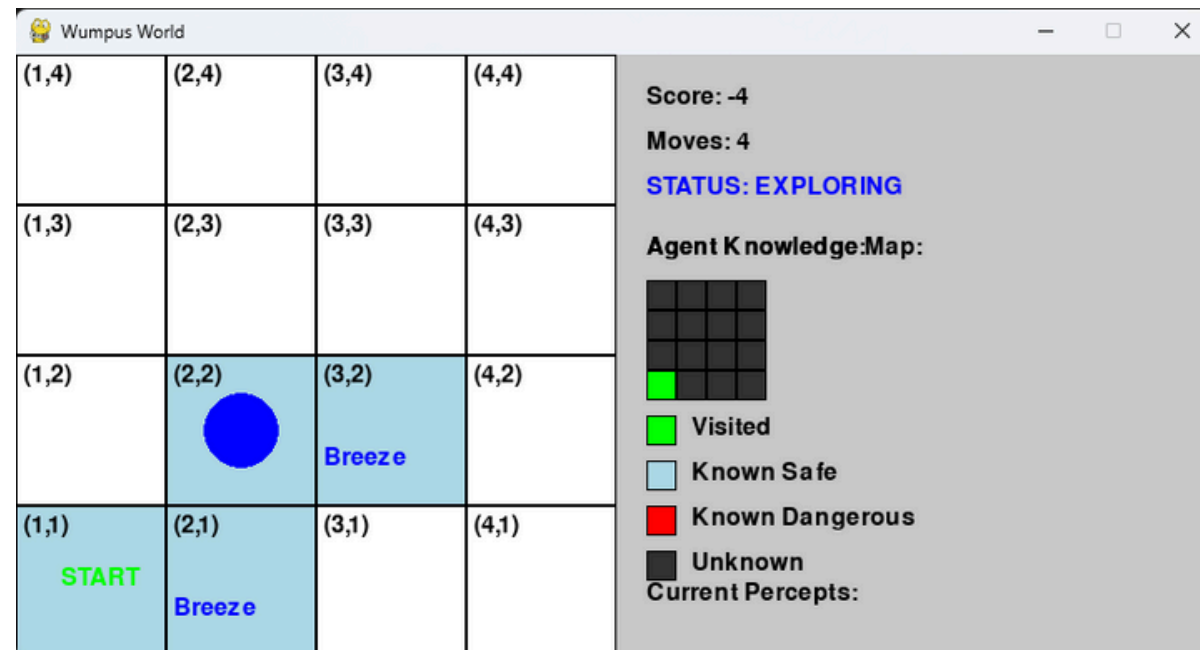
Screenshots



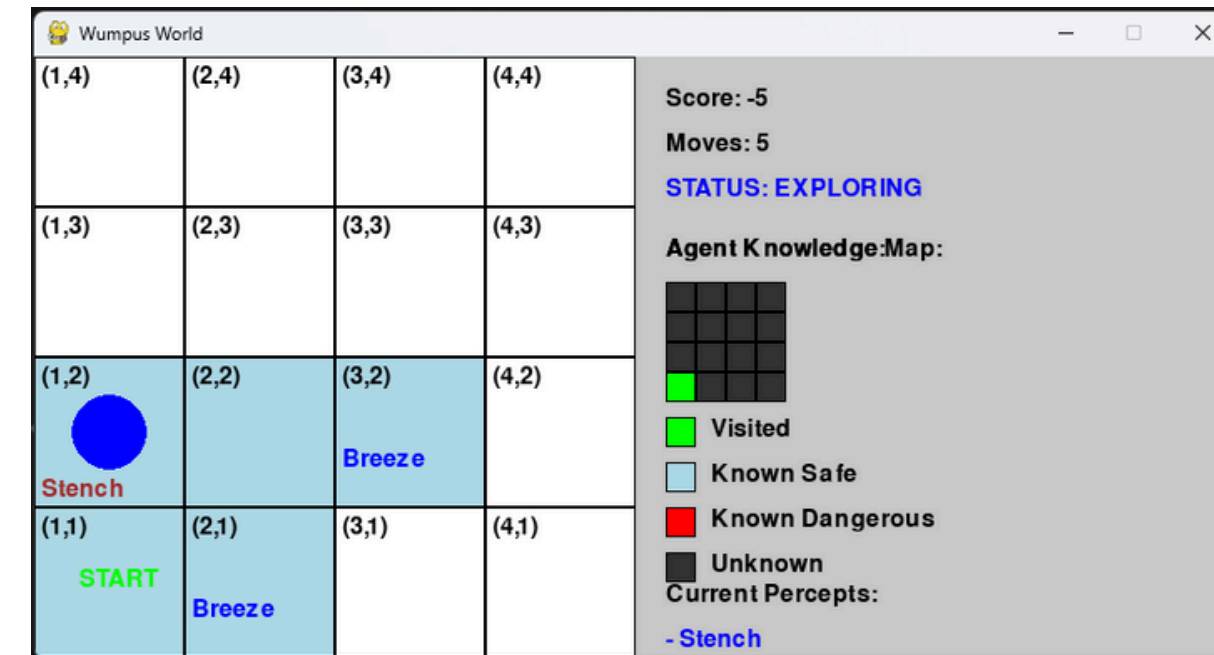
- Start Screen



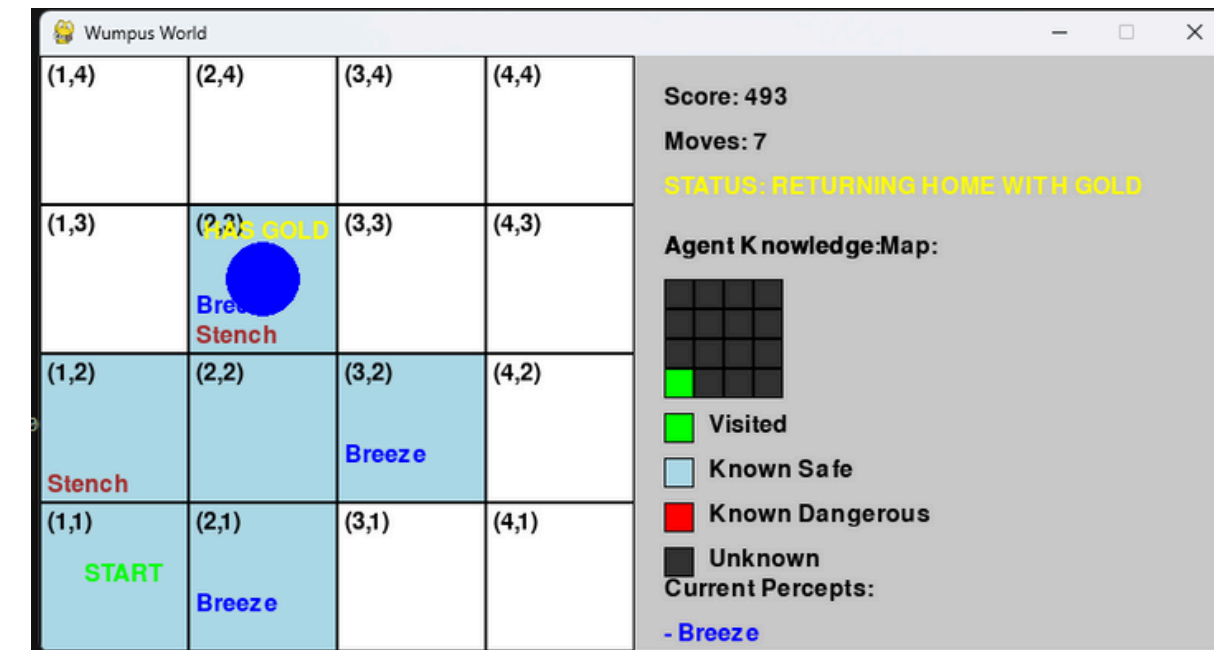
- Breeze Sensed



- Stench Sensed



- GOLD Grabbed



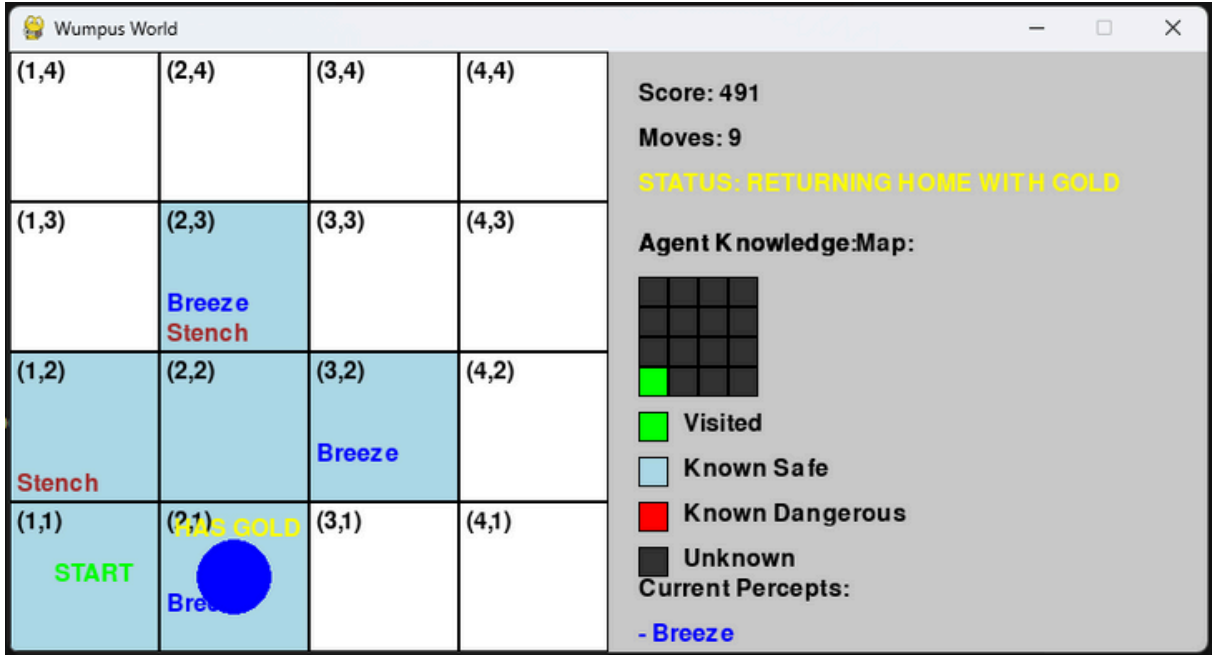
Results



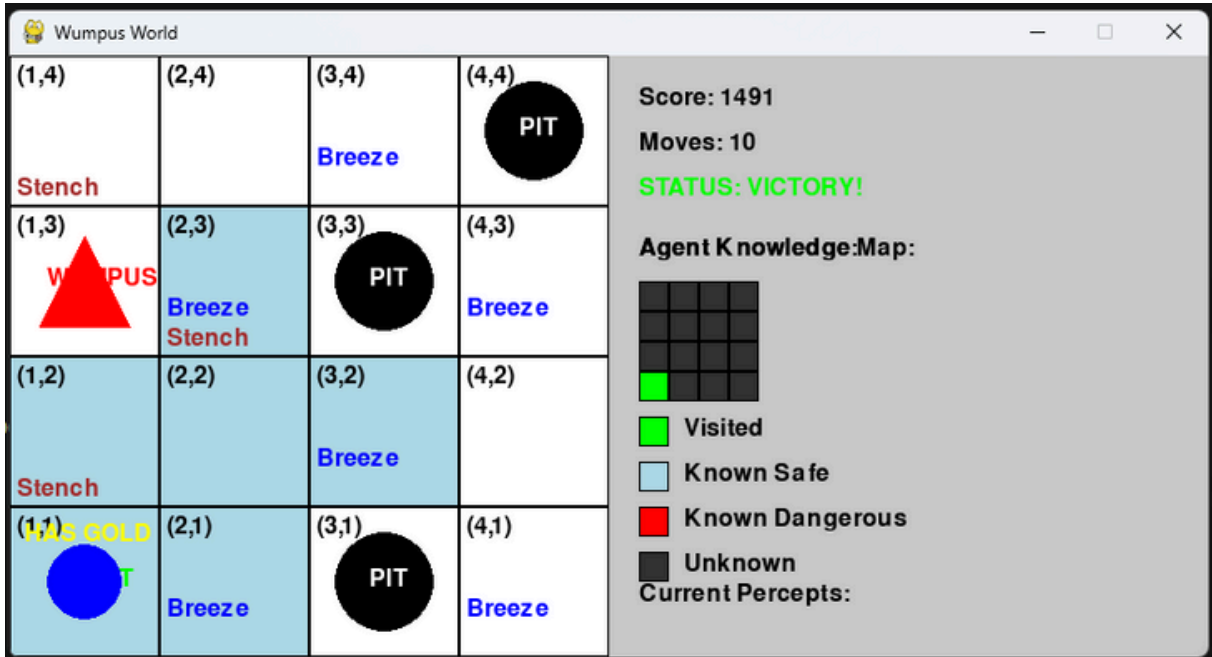
Screenshots



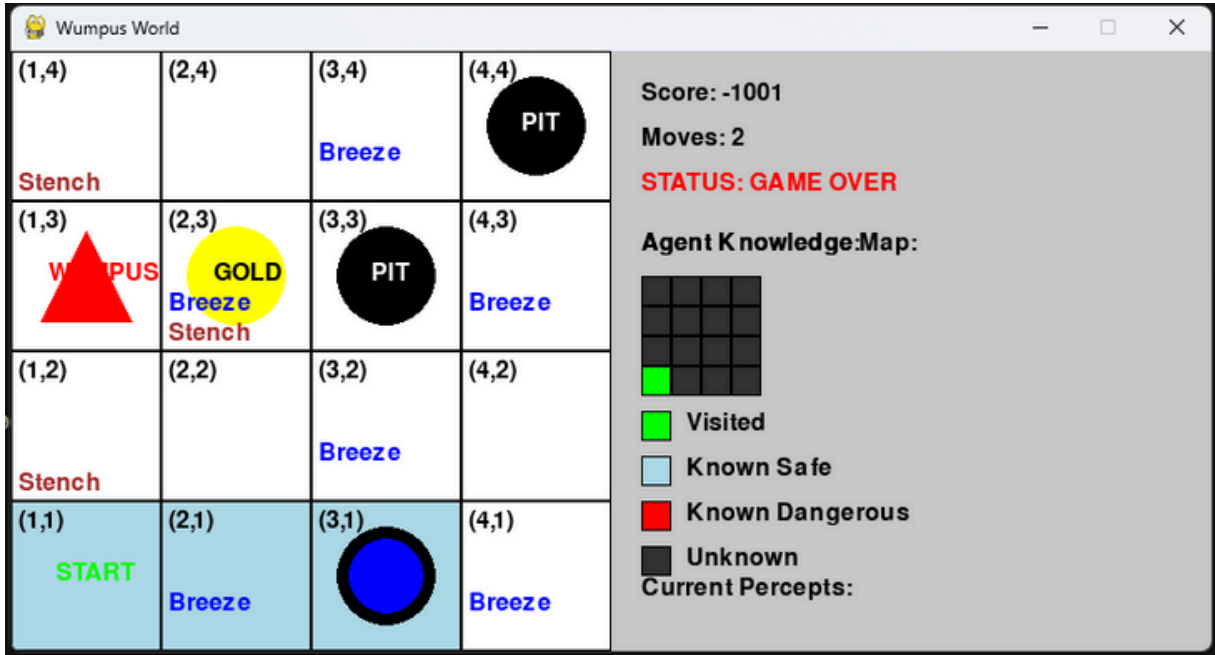
- Back-Tracking



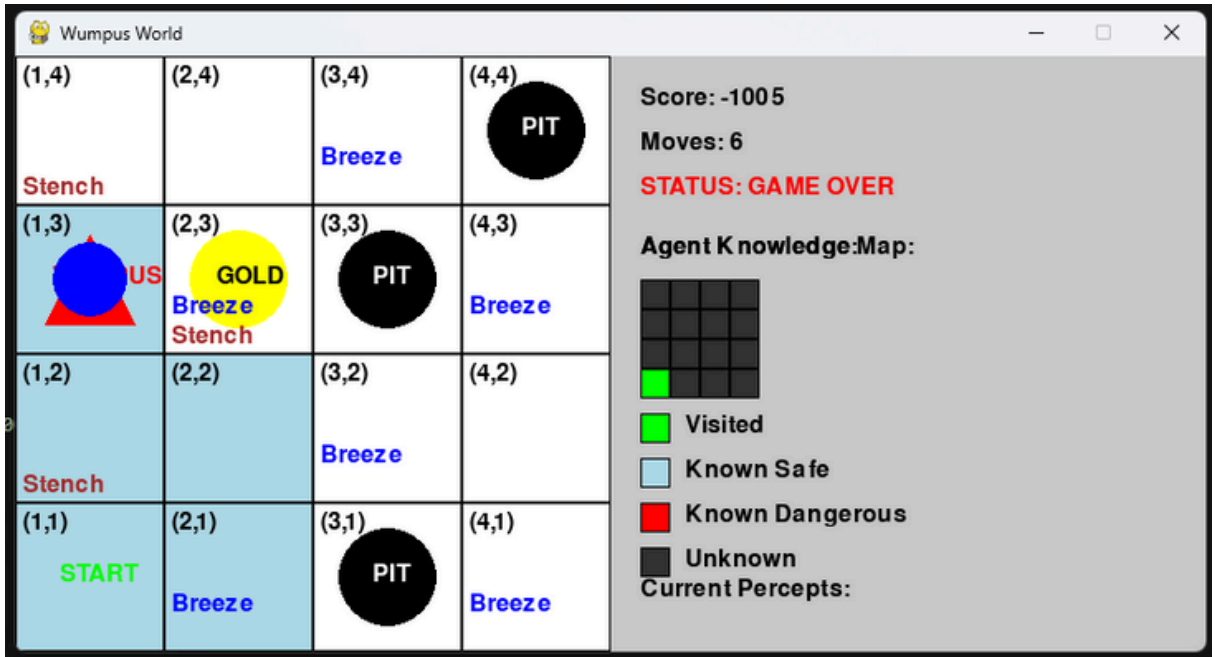
- Game Won



- Fall in PIT (Game Over)



- Eaten by Wumpus (Game Over)



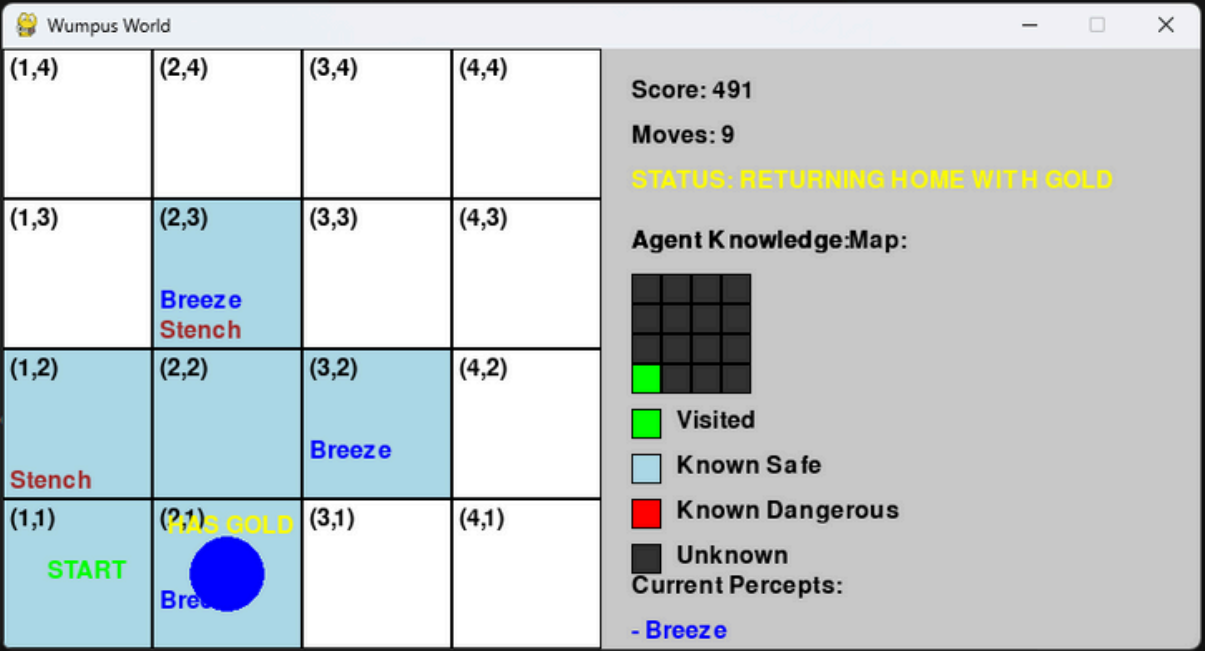
Results



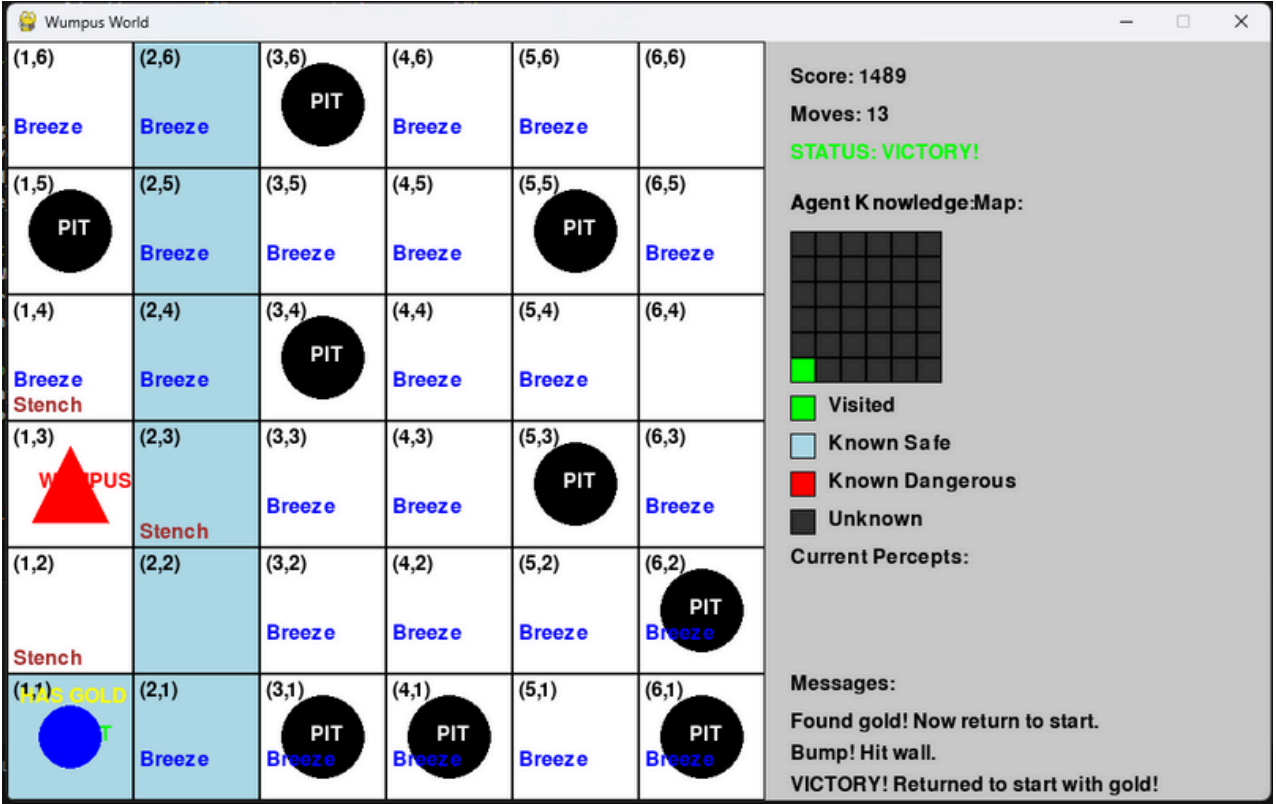
Screenshots



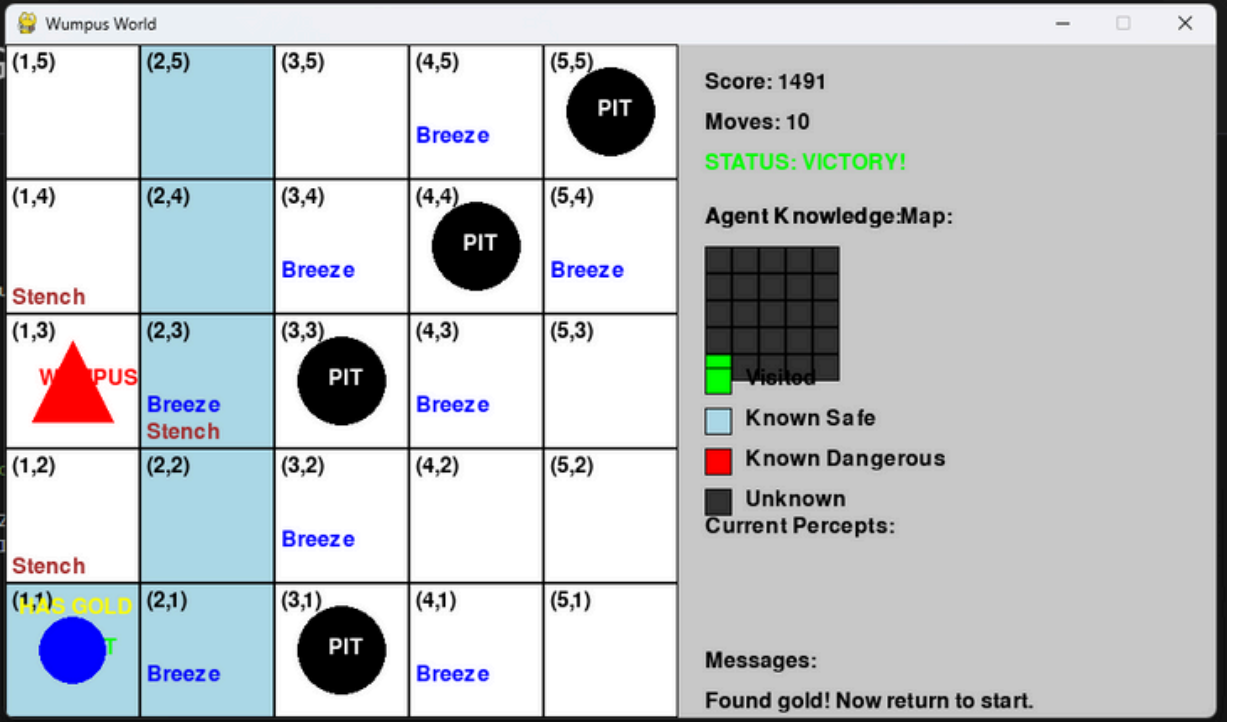
Different Levels



LEVEL 1 : 4 X 4



LEVEL 3 : 6 X 6



LEVEL 2 : 5 X 5

 **Many More!**
Coming Soon!!

Results



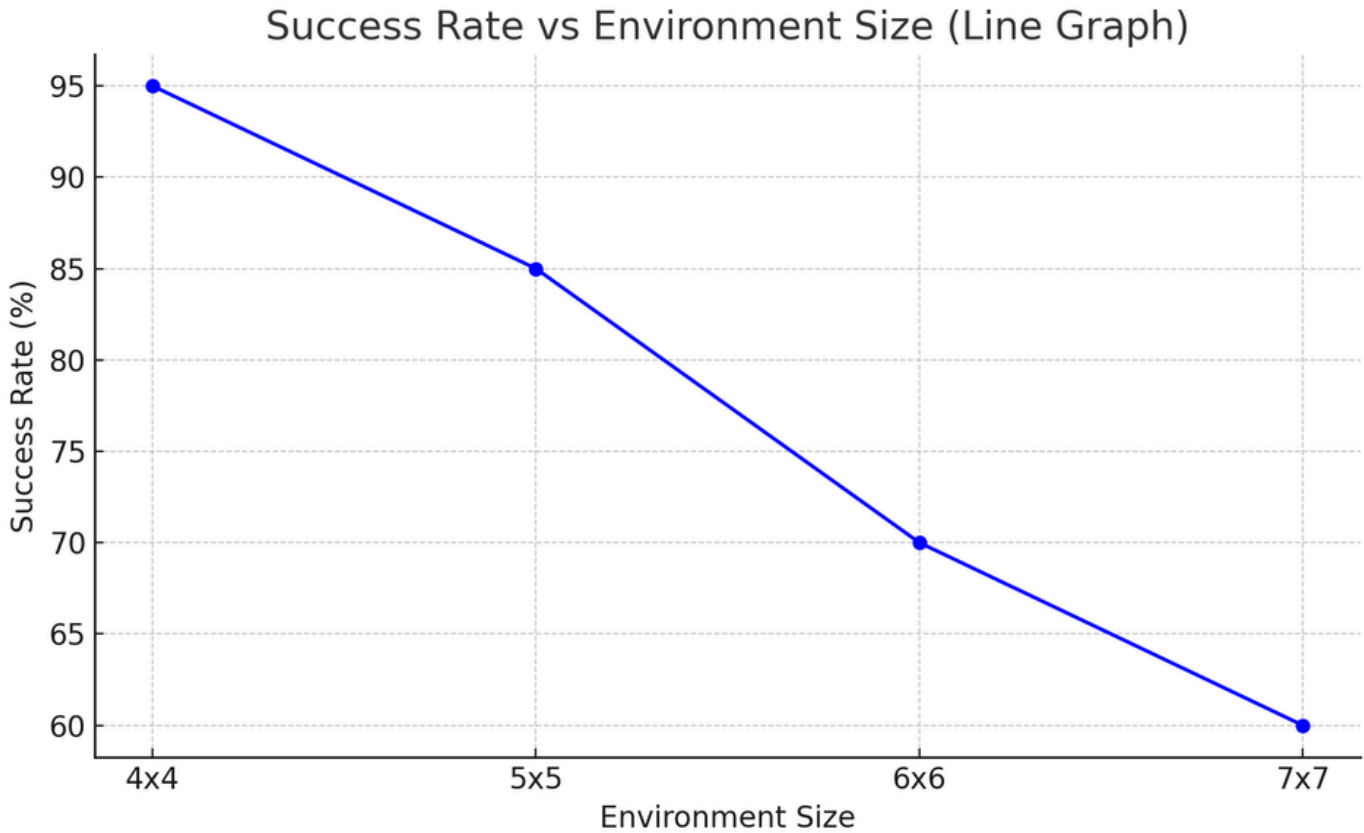
Comparison graphs



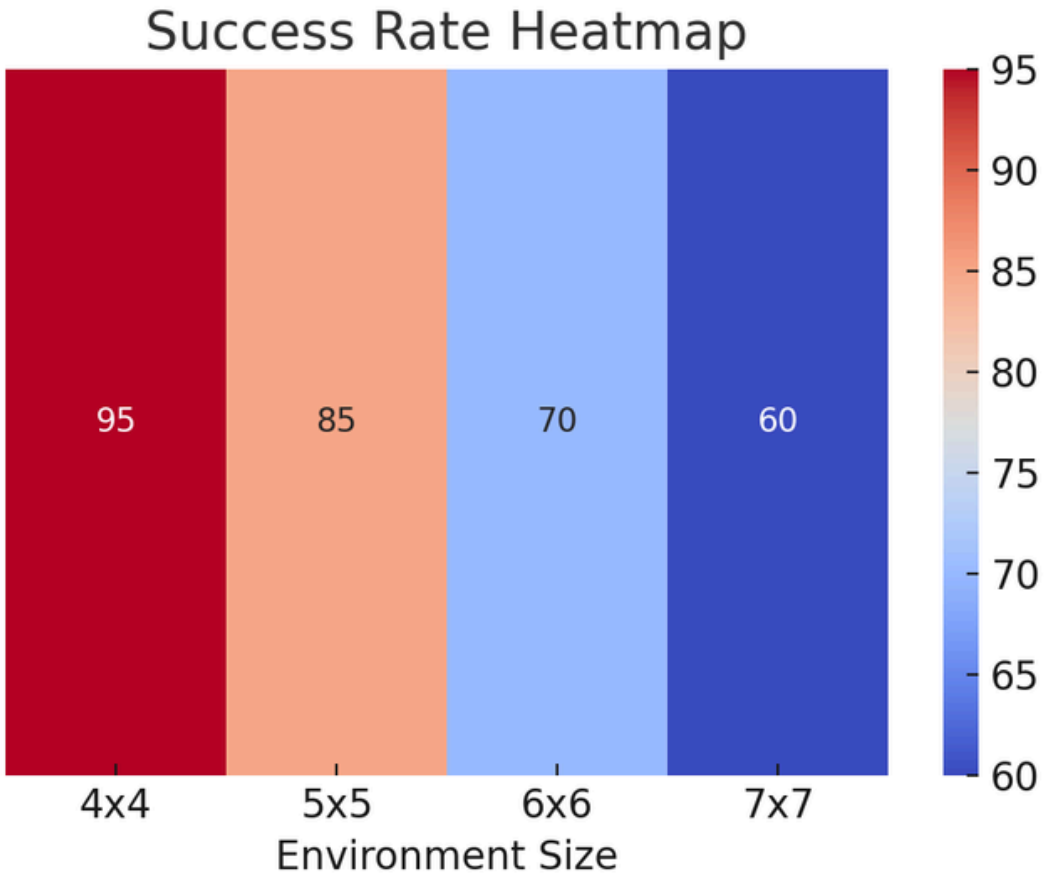
Environment Size	Success Rate (%)
4x4	95%
5x5	85%
6x6	70%
7x7	60%



Line Graph



Heat Map/ Confusion Graph



Results



DEMO VIDEO



Click To Check the Demo Video

File Edit Selection View Go Run Terminal Help

WUMPUS

venv (Python 3.12.0)

EXPLORER

WUMPUS

venv

Flowchart_wum...

README.md

wumpus.ipynb

wumpus.ipynb

README.md

wumpus.ipynb

WUMPUS WORLD AI SOLVER

import pygame

Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline

Note: you may need to restart the kernel to use updated packages.

```
import pygame
import random
import time
import math
from enum import Enum, auto
from collections import deque
import heapq

# Initialize pygame
pygame.init()

# Constants
GRID_SIZE = 4
CELL_SIZE = 100
WIDTH = GRID_SIZE * CELL_SIZE + 400 # Extra space for info panel
HEIGHT = GRID_SIZE * CELL_SIZE
FPS = 30

# Colors
WHITE = (255, 255, 255)
BLACK = (0, 0, 0)
GRAY = (200, 200, 200)
RED = (255, 0, 0)
GREEN = (0, 255, 0)
BLUE = (0, 0, 255)
YELLOW = (255, 255, 0)
BROWN = (165, 42, 42)
DARK_GRAY = (50, 50, 50)
LIGHT_BLUE = (173, 216, 230)

# Direction enum
class Direction(Enum):
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

JUPYTER

No problems have been detected in the workspace.

OUTLINE

TIMELINE

main*

0 0 0

Reactivating terminals...

Wumpus World

(1,4)	(2,4)	(3,4)	(4,4)
Stench		Breeze	PIT
(1,3)	(2,3)	(3,3)	(4,3)
US	GOLD	PIT	Breeze
(1,2)	(2,2)	(3,2)	(4,2)
Stench		Breeze	
(1,1)	(2,1)	(3,1)	(4,1)
START	Breeze	PIT	Breeze

Score: -1005

Moves: 6

STATUS: GAME OVER

Agent Knowledge Map:

Visited

Known Safe

Known Dangerous

Unknown

Current Percepts:

Filter (e.g. text, **/*.ts, !**/node_modules/**)

Spaces: 4

Cell 3 of 4

ENG IN

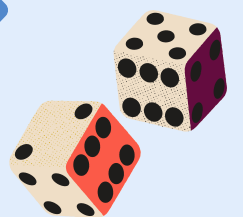
4:57 PM

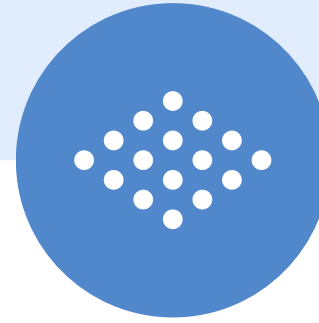
4/28/2025



Conclusion

The project successfully demonstrated an intelligent agent navigating the Wumpus World, avoiding hazards while retrieving gold. Performance was affected by grid size and hazard complexity, with larger grids leading to lower success rates. Future improvements include using probabilistic reasoning and machine learning to enhance decision-making and agent adaptability in dynamic environments.





THANK YOU!



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Soubhik Sadhu (RA2311026010122)